

More Simple Statistics

Document Number: P2681R0
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Date: August 15, 2022 (mailing)
Project: ISO JTC1/SC22/WG21: Programming Language C++
Audience: SG6, SG19, LEWG

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0 Revision History

R####R1

- TBA

1 Introduction

This document proposes an extension to the C++ library, to support more **simple statistics**.

2 Motivation and Scope

Simple statistical functions, **five** of which are proposed as part of P1708R6 [1], frequently arise in **scientific** and **industrial**, as well as **general**, applications. These functions do exist in Python [2], the foremost competitor to C++ in the area of **machine learning**, along with Calc [3], Excel [4], Julia [5], MATLAB [6], PHP [7], R [8], Rust [9], SAS [10] and SPSS [11]. Further need for such functions has been identified as part of **SG19** (machine learning) [12].

This is not the first proposal to move statistics in C++. In 2004, a number of statistical distributions were proposed in [13]. Additional distributions followed in 2006 [14]. Statistical distributions ultimately appeared in the C++11 standard [15]. Distributions, along with statistical tests, are also found in Boost [16]. A series of special mathematical functions later followed as part of the C++17 standard [17]. A C library, GNU Scientific Library [18], further includes support for statistics, special functions and histograms.

Five more statistics are defined in this proposal. As in P1708R6 [1], both freestanding **functions** and accumulator **objects** are proposed. Like existing entities of the (C++) standard library, this proposal specifies only the interface of functions and objects, meaning that a variety of implementations are possible. This enables a vendor to favor accuracy [19] over performance for instance.

2.1 Univariate Statistics

Univariate [20] statistics, which include those of P1708R6 [1], are computed over the (single set of) values $x_1, x_2, \dots, x_n$ ($n \geq 1$). The following **two** univariate statistics are put forward in this proposal.

2.1.1 Percentile

A *percentile* [21] is defined as the value p , in the range $(0, 1)$, below which $100p\%$ of the (**sorted**) values fall. As an example, the $p = 0.5$ percentile, known as the *median* [22], divides the values into two intervals (halves) of equal size. It is therefore the “middle” value. Both **even** and **odd** n [23] must be considered in the case of discrete data. At this point, “[t]here is no standard function for a weighted percentile” [24]. However, for weights $w_1, w_2, \dots, w_n$, such that

$$\sum_{i=1}^n w_i = 1, \tag{1}$$

the *weighted median* is defined as the x_i for which

$$\sum_{i=1}^{k-1} w_i \leq \frac{1}{2} \tag{2}$$

and

$$\sum_{i=k+1}^n w_i \leq \frac{1}{2}. \tag{3}$$

A percentile (of **sorted** values) can be found in **linear** time.

2.1.2 Mode

The *mode* [22] is defined as the (perhaps not unique) **most frequent** value of the values. The *weighted mode* [25] is defined as the (perhaps not unique) value with the highest total weight. The mode can be found in **linear** time by comparing adjacent (**sorted**) values.

2.2 Bivariate Statistics

Bivariate [20] statistics are computed over the (two sets of) values $x_1, x_2, \dots, x_n$ and $y_1, y_2, \dots, y_n$. The following **three** bivariate statistics are put forward in this proposal.

2.2.1 Covariance

The **population covariance** [26] is a measure of the **joint variability** of the values. It is defined as

$$\sigma_{xy} = \frac{1}{n} \sum_{i=1}^n (x_i - \mu_x)(y_i - \mu_y), \quad (4)$$

where μ_x and μ_y are the (arithmetic) **population means** [22] of the values $x_1, x_2, \dots, x_n$ and $y_1, y_2, \dots, y_n$, respectively. The **sample** [26] **covariance** is defined as

$$s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}), \quad (5)$$

where $\bar{x}$ and $\bar{y}$ are the **sample means** [22] of the values. The **weighted population** and **sample covariance** [27] is defined as

$$\frac{\sum_{i=1}^n w_i (x_i - \hat{x})(y_i - \hat{y})}{\sum_{i=1}^n w_i}, \quad (6)$$

where $\hat{x}$ and $\hat{y}$ are the **weighted population** or **sample means** [28, 29].

2.2.2 Correlation

The **Pearson population correlation coefficient** [30], in the range $[-1, 1]$, is a measure of the **linear association** of the values. It is defined as

$$\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}, \quad (7)$$

where σ_x and σ_y are the **population standard deviations** [22] of the values $x_1, x_2, \dots, x_n$ and $y_1, y_2, \dots, y_n$, respectively. The **Pearson sample correlation coefficient** [30] is defined as

$$r_{xy} = \frac{s_{xy}}{s_x s_y}, \quad (8)$$

where s_x and s_y are the **sample standard deviations** [22] of the values. The **weighted Pearson population** and **sample correlation coefficient** [27] is defined as

$$\frac{\sum_{i=1}^n w_i (x_i - \hat{x})(y_i - \hat{y})}{\sum_{i=1}^n w_i} \sqrt{\frac{\sum_{i=1}^n w_i (x_i - \hat{x})^2}{\sum_{i=1}^n w_i} \frac{\sum_{i=1}^n w_i (y_i - \hat{y})^2}{\sum_{i=1}^n w_i}}. \quad (9)$$

2.2.3 Linear Regression

Linear regression [30, 31] is a **straight-line** model of the values. The **y-intercept** of the model is defined as

$$\hat{y} - \frac{\hat{x} \sum_{i=1}^n (x_i - \hat{x})(y_i - \hat{y})}{\sum_{i=1}^n (x_i - \hat{x})^2} \quad (10)$$

and the **slope** is defined as

$$\frac{\sum_{i=1}^n (x_i - \hat{x})(y_i - \hat{y})}{\sum_{i=1}^n (x_i - \hat{x})^2}, \quad (11)$$

where $\bar{x}$ and $\bar{y}$ are the **population** or **sample** means. The **y-intercept** of the model in the case of *weighted linear regression* [32] is defined as

$$\hat{y} - \frac{\hat{x} \sum_{i=1}^n w_i (x_i - \hat{x})(y_i - \hat{y})}{\sum_{i=1}^n w_i (x_i - \hat{x})^2} \quad (12)$$

and the **slope** is defined as

$$\frac{\sum_{i=1}^n w_i (x_i - \hat{x})(y_i - \hat{y})}{\sum_{i=1}^n w_i (x_i - \hat{x})^2}. \quad (13)$$

3 Impact on the Standard

This proposal is a pure **library** extension.

4 Design Decisions

The discussions of the following sections address the concerns that have been raised in regards to this proposal.

4.1 Naming

Unlike the statistics of P1708R6 [1], the **percentiles** and **modes** of this proposal require that ranges be **sorted**. To ensure that users are aware of this fact, the names of the percentile and mode functions (and mode objects) include the word “sorted”. Initially, it was proposed that the names of these functions be **prepended** by `sorted_`, which would have resulted in names such as `sorted_percentiles`. It is believed though that these names would be **misleading**, as they would suggest that the statistics (percentiles in this example) are sorted, rather than the values of the ranges. Thus, `_of_sorted` is instead **appended** to the names of these functions, resulting in less confusing names, such as `percentiles_of_sorted`.

4.2 Percentile Accumulator Objects

Though percentile **functions** are proposed, percentile **objects** are **not** proposed. An accumulator object will most likely maintain a counter corresponding to the position of the current value (of a range). This object would therefore need to perform a (costly) conditional test for each value of a range in order to determine if the counter corresponds to a particular percentile. This repeated testing results in an unacceptably high run-time complexity. Functions, by comparison, allow one to (quickly) advance an iterator to the computed position of a percentile without having to visit (any) intermediate values. Even better, because the determination of a percentile requires that a range be **sorted**, a `random_access_range` will often be used, for which there is direct access (to a computed position).

4.3 Quantile

A *quantile* [33] is defined as the value x_i that divides the (**sorted**) values into intervals of **equal** size. As an example, the 4-quantiles, or *quartiles* [34], divide the values into four intervals, bounded by the $1/4 = 0.25$, $2/4 = 0.5$ and $3/4 = 0.75$ percentiles. A user wishing to compute the 4-quantiles can readily do so via a percentile function, perhaps

```
std::percentiles_of_sorted(data, std::vector<double>{0.25, 0.5, 0.75}, results.begin());
```

Some might suggest that quantiles be provided as a convenience function, perhaps of the form

```
std::quantiles_of_sorted(data, 4, results.begin());
```

Unfortunately, if the desired number of quantiles is not known at run-time, then a **container**, such as a list or vector, must be passed to any function. Along with a container, a user would logically expect to have the ability to specify an **allocator**, further complicating the situation. For these reasons, quantiles are **not** provided.

5 Technical Specifications

The templates of the classes and functions of the percentile and mode statistics specified in this section are defined for any type. The templates of the covariance, correlation and linear regression statistics specified in this section are defined for each of the arithmetic types, except for **bool**. In the case of the covariance, correlation and linear regression statistics, the effect of instantiating the templates for any other type is unspecified. Parallel function overloads follow the requirements of [algorithms.parallel].

5.1 Header <stats> synopsis [stats.syn]

```
#include <execution>

namespace std {

// ... existing accumulator objects ...

// mode accumulator class templates
template<class T>
class mode_of_sorted_accumulator;

template<class T, std::weakly_incrementable O>
requires std::convertible_to<T, std::iter_value_t<O>>
class modes_of_sorted_accumulator;

template<class T, class W>
class weighted_mode_of_sorted_accumulator;

template<class T, class W, std::weakly_incrementable O>
requires std::convertible_to<T, std::iter_value_t<O>>
class weighted_modes_of_sorted_accumulator;

// ... existing accumulator objects ...

// covariance accumulator class templates
template<class T1, class T2>
requires convertible_to<T1, T2> || convertible_to<T2, T1>
class covariance_accumulator;

template<class T1, class T2, class W>
requires convertible_to<T1, T2> || convertible_to<T2, T1>
class weighted_population_covariance_accumulator;

template<class T1, class T2, class W>
requires convertible_to<T1, T2> || convertible_to<T2, T1>
class weighted_sample_covariance_accumulator;

// correlation accumulator class templates
template<class T1, class T2>
requires convertible_to<T1, T2> || convertible_to<T2, T1>
class correlation_accumulator;

template<class T1, class T2, class W>
requires convertible_to<T1, T2> || convertible_to<T2, T1>
class weighted_population_correlation_accumulator;

template<class T1, class T2, class W>
requires convertible_to<T1, T2> || convertible_to<T2, T1>
class weighted_sample_correlation_accumulator;
```

```

// linear regression accumulator
template<class T>
struct linear_regression_result
{
    T intercept, slope;
};

template<class T1, class T2>
requires convertible_to<T1, T2> || convertible_to<T2, T1>
class linear_regression_accumulator;

template<class T1, class T2, class W>
requires convertible_to<T1, T2> || convertible_to<T2, T1>
class weighted_population_linear_regression_accumulator;

template<class T1, class T2, class W>
requires convertible_to<T1, T2> || convertible_to<T2, T1>
class weighted_sample_linear_regression_accumulator;

// accumulator objects accumulation functions

// ... existing functions ...

template<ranges::input_range R1, ranges::input_range R2, class ...Accumulators>
constexpr void stats_accumulate(ranges::zip_view<R1, R2>&& r, Accumulators&& ... acc);

template<ranges::input_range R1,
    ranges::input_range R2,
    ranges::input_range W,
    class ...Accumulators>
constexpr void weighted_stats_accumulate(
    ranges::zip_view<R1, R2>&& r, W&& w, Accumulators&& ... acc);

template<class ExecutionPolicy,
    ranges::input_range R1,
    ranges::input_range R2,
    class ...Accumulators>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>>
void stats_accumulate(
    ExecutionPolicy&& policy, ranges::zip_view<R1, R2>&& r, Accumulators&& ... acc);

template<class ExecutionPolicy,
    ranges::input_range R1,
    ranges::input_range R2,
    ranges::input_range W,
    class ...Accumulators>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>>
void weighted_stats_accumulate(
    ExecutionPolicy&& policy, ranges::zip_view<R1, R2>&& r, W&& w, Accumulators&& ... acc);

// ... existing functions ...

// freestanding percentile functions
template<ranges::sized_range R>
constexpr auto percentile_of_sorted(R&& r, double p) ->
    ranges::subrange<ranges::iterator_t<R>>;

template<ranges::sized_range R, ranges::input_range P, std::weakly_incrementable O>

```

```

/* ... requires ... */

constexpr auto percentiles_of_sorted(R&& r, P&& p, O it) -> O;

template<ranges::sized_range R>
constexpr auto median_of_sorted(R&& r) -> ranges::subrange<ranges::iterator_t<R>>;

template<ranges::sized_range R, ranges::input_range W>
constexpr auto weighted_median_of_sorted(R&& r, W&& w) ->
    ranges::subrange<ranges::iterator_t<R>>;

template<class ExecutionPolicy, ranges::sized_range R>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>>
constexpr auto percentile_of_sorted(ExecutionPolicy&& policy, R&& r, double p) ->
    ranges::subrange<ranges::iterator_t<R>>;

template<class ExecutionPolicy,
    ranges::sized_range R,
    ranges::input_range P,
    std::weakly_incremtable O>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>>

/* ... && ... */

constexpr auto percentiles_of_sorted(ExecutionPolicy&& policy, R&& r, P&& p, O it) -> O;

template<class ExecutionPolicy, ranges::sized_range R>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>>
constexpr auto median_of_sorted(ExecutionPolicy&& policy, R&& r) ->
    ranges::subrange<ranges::iterator_t<R>>;

template<class ExecutionPolicy, ranges::sized_range R, ranges::input_range W>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>>
constexpr auto weighted_median_of_sorted(ExecutionPolicy&& policy, R&& r, W&& w) ->
    ranges::subrange<ranges::iterator_t<R>>;

// freestanding mode functions
template<ranges::input_range R>
constexpr auto mode_of_sorted(R&& r) -> ranges::iterator_t<R>::value_type;

template<ranges::input_range R, ranges::input_range W>
constexpr auto weighted_mode_of_sorted(R&& r, W&& w) -> ranges::iterator_t<R>::value_type;

template<ranges::input_range R, std::weakly_incremtable O>
requires std::indirectly_copyable<ranges::iterator_t<R>, O>
constexpr auto modes_of_sorted(R&& r, size_t n, O it) -> O;

template<ranges::input_range R, ranges::input_range W, std::weakly_incremtable O>
requires std::indirectly_copyable<ranges::iterator_t<R>, O>
constexpr auto weighted_modes_of_sorted(R&& r, W&& w, size_t n, O it) -> O;

template<class ExecutionPolicy, ranges::input_range R>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>>
constexpr auto mode_of_sorted(ExecutionPolicy&& policy, R&& r) ->
    ranges::iterator_t<R>::value_type;

template<class ExecutionPolicy, ranges::input_range R, ranges::input_range W>

```



```

requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>>
constexpr auto weighted_mode_of_sorted(ExecutionPolicy&& policy, R&& r, W&& w) ->
    ranges::iterator_t<R>::value_type;

template<class ExecutionPolicy, ranges::input_range R, std::weakly_increamentable O>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>> &&
    std::indirectly_copyable<ranges::iterator_t<R>, O>
constexpr auto modes_of_sorted(ExecutionPolicy&& policy, R&& r, size_t n, O it) -> O;

template<class ExecutionPolicy,
    ranges::input_range R,
    ranges::input_range W,
    std::weakly_increamentable O>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>> &&
    std::indirectly_copyable<ranges::iterator_t<R>, O>
constexpr auto weighted_modes_of_sorted(
    ExecutionPolicy&& policy, R&& r, W&& w, size_t n, O it) -> O;

// ... existing functions ...

// freestanding covariance functions
template<ranges::input_range R1, ranges::input_range R2>
requires convertible_to<R1, R2> || convertible_to<R2, R1>
constexpr auto covariance(ranges::zip_view<R1, R2>&& r,
    typename std::conditional<
        std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
        std::iter_value_t<R1>,
        std::iter_value_t<R2>>::type ddof) ->
    std::conditional<
        std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
        std::iter_value_t<R1>,
        std::iter_value_t<R2>>::type;

template<ranges::input_range R1, ranges::input_range R2, ranges::input_range W>
requires convertible_to<R1, R2> || convertible_to<R2, R1>
constexpr auto weighted_population_covariance(ranges::zip_view<R1, R2>&& r, W&& w) ->
    std::conditional<std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
        std::iter_value_t<R1>,
        std::iter_value_t<R2>>::type;

template<ranges::input_range R1, ranges::input_range R2, ranges::input_range W>
requires convertible_to<R1, R2> || convertible_to<R2, R1>
constexpr auto weighted_sample_covariance(ranges::zip_view<R1, R2>&& r, W&& w) ->
    std::conditional<std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
        std::iter_value_t<R1>,
        std::iter_value_t<R2>>::type;

template<class ExecutionPolicy, ranges::input_range R1, ranges::input_range R2>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>> &&
    (convertible_to<R1, R2> || convertible_to<R2, R1>)
constexpr auto covariance(ExecutionPolicy&& policy,
    ranges::zip_view<R1, R2>&& r,
    typename std::conditional<std::is_convertible_v<
        std::iter_value_t<R1>, std::iter_value_t<R2>>,
        std::iter_value_t<R1>,
        std::iter_value_t<R2>>::type ddof) ->
    std::conditional<
        std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,

```

```

        std::iter_value_t<R1>,
        std::iter_value_t<R2>>::type;

template<class ExecutionPolicy,
        ranges::input_range R1,
        ranges::input_range R2,
        ranges::input_range W>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>> &&
        (convertible_to<R1, R2> || convertible_to<R2, R1>)
constexpr auto weighted_population_covariance(
    ExecutionPolicy&& policy, ranges::zip_view<R1, R2>&& r, W&& w) ->
    std::conditional<std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
        std::iter_value_t<R1>,
        std::iter_value_t<R2>>::type;

template<class ExecutionPolicy,
        ranges::input_range R1,
        ranges::input_range R2,
        ranges::input_range W>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>> &&
        (convertible_to<R1, R2> || convertible_to<R2, R1>)
constexpr auto weighted_sample_covariance(
    ExecutionPolicy&& policy, ranges::zip_view<R1, R2>&& r, W&& w) ->
    std::conditional<std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
        std::iter_value_t<R1>,
        std::iter_value_t<R2>>::type;

// freestanding correlation functions
template<ranges::input_range R1, ranges::input_range R2>
requires convertible_to<R1, R2> || convertible_to<R2, R1>
constexpr auto correlation(ranges::zip_view<R1, R2>&& r,
    typename std::conditional<
        std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
        std::iter_value_t<R1>,
        std::iter_value_t<R2>>::type ddof) ->
    std::conditional<
        std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
        std::iter_value_t<R1>,
        std::iter_value_t<R2>>::type;

template<ranges::input_range R1, ranges::input_range R2, ranges::input_range W>
requires convertible_to<R1, R2> || convertible_to<R2, R1>
constexpr auto weighted_population_correlation(ranges::zip_view<R1, R2>&& r, W&& w) ->
    std::conditional<std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
        std::iter_value_t<R1>,
        std::iter_value_t<R2>>::type;

template<ranges::input_range R1, ranges::input_range R2, ranges::input_range W>
requires convertible_to<R1, R2> || convertible_to<R2, R1>
constexpr auto weighted_sample_correlation(ranges::zip_view<R1, R2>&& r, W&& w) ->
    std::conditional<std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
        std::iter_value_t<R1>,
        std::iter_value_t<R2>>::type;

template<class ExecutionPolicy, ranges::input_range R1, ranges::input_range R2>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>> &&
        (convertible_to<R1, R2> || convertible_to<R2, R1>)
constexpr auto correlation(ExecutionPolicy&& policy,

```

```

ranges::zip_view<R1, R2>&& r,
typename std::conditional<std::is_convertible_v<
    std::iter_value_t<R1>, std::iter_value_t<R2>>,
    std::iter_value_t<R1>,
    std::iter_value_t<R2>>::type ddof) ->
    std::conditional<
        std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
        std::iter_value_t<R1>,
        std::iter_value_t<R2>>::type;

template<class ExecutionPolicy,
    ranges::input_range R1,
    ranges::input_range R2,
    ranges::input_range W>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>> &&
    (convertible_to<R1, R2> || convertible_to<R2, R1>)
constexpr auto weighted_population_correlation(
    ExecutionPolicy&& policy, ranges::zip_view<R1, R2>&& r, W&& w) ->
    std::conditional<std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
        std::iter_value_t<R1>,
        std::iter_value_t<R2>>::type;

template<class ExecutionPolicy,
    ranges::input_range R1,
    ranges::input_range R2,
    ranges::input_range W>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>> &&
    (convertible_to<R1, R2> || convertible_to<R2, R1>)
constexpr auto weighted_sample_correlation(
    ExecutionPolicy&& policy, ranges::zip_view<R1, R2>&& r, W&& w) ->
    std::conditional<std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
        std::iter_value_t<R1>,
        std::iter_value_t<R2>>::type;

// freestanding linear regression functions
template<ranges::input_range R1, ranges::input_range R2>
requires convertible_to<R1, R2> || convertible_to<R2, R1>
constexpr auto linear_regression(ranges::zip_view<R1, R2>&& r) ->
    linear_regression_result<
        typename std::conditional<
            std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
            std::iter_value_t<R1>,
            std::iter_value_t<R2>>::type>;

template<ranges::input_range R1, ranges::input_range R2, ranges::input_range W>
requires convertible_to<R1, R2> || convertible_to<R2, R1>
constexpr auto weighted_population_linear_regression(
    ranges::zip_view<R1, R2>&& r, W&& w) ->
    linear_regression_result<
        typename std::conditional<
            std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
            std::iter_value_t<R1>,
            std::iter_value_t<R2>>::type>;

template<ranges::input_range R1, ranges::input_range R2, ranges::input_range W>
requires convertible_to<R1, R2> || convertible_to<R2, R1>
constexpr auto weighted_sample_linear_regression(
    ranges::zip_view<R1, R2>&& r, W&& w) ->

```

```

linear_regression_result<
    typename std::conditional<
        std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
        std::iter_value_t<R1>,
        std::iter_value_t<R2>>::type>;

template<class ExecutionPolicy, ranges::input_range R1, ranges::input_range R2>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>> &&
    (convertible_to<R1, R2> || convertible_to<R2, R1>)
constexpr auto linear_regression(
    ExecutionPolicy&& policy, ranges::zip_view<R1, R2>&& r) ->
    linear_regression_result<
        typename std::conditional<
            std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
            std::iter_value_t<R1>,
            std::iter_value_t<R2>>::type>;

template<class ExecutionPolicy,
    ranges::input_range R1,
    ranges::input_range R2,
    ranges::input_range W>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>> &&
    (convertible_to<R1, R2> || convertible_to<R2, R1>)
auto weighted_population_linear_regression(
    ExecutionPolicy&& policy, ranges::zip_view<R1, R2>&& r, W&& w) ->
    linear_regression_result<
        typename std::conditional<
            std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
            std::iter_value_t<R1>,
            std::iter_value_t<R2>>::type>;

template<class ExecutionPolicy,
    ranges::input_range R1,
    ranges::input_range R2,
    ranges::input_range W>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>> &&
    (convertible_to<R1, R2> || convertible_to<R2, R1>)
auto weighted_sample_linear_regression(
    ExecutionPolicy&& policy, ranges::zip_view<R1, R2>&& r, W&& w) ->
    linear_regression_result<
        typename std::conditional<
            std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
            std::iter_value_t<R1>,
            std::iter_value_t<R2>>::type>;
}

```

5.2 Accumulator Objects

The accumulator objects specified in this section are trivially copyable. If any of the values of x , y or w of the function **operator** () specified in this section is a NaN, ∞ or $-\infty$, or overflow or underflow occurs, which might even occur in the case of finite ranges of values, the function value returns an unspecified value.

5.2.1 Mode Accumulator Class Templates

```

template<class T>
class mode_of_sorted_accumulator
{
public:

```

```

explicit mode_of_sorted_accumulator() noexcept;
constexpr void operator() (const T& x);
constexpr auto value() -> T;
};

template<class T, std::weakly_increamentable O>
requires std::convertible_to<T, std::iter_value_t<O>>
class modes_of_sorted_accumulator
{
public:
    explicit modes_of_sorted_accumulator(size_t n, O it) noexcept;
    constexpr void operator() (const T& x);
    constexpr auto value() -> O;
};

template<class T, class W>
class weighted_mode_of_sorted_accumulator
{
public:
    explicit weighted_mode_of_sorted_accumulator() noexcept;
    constexpr void operator() (const T& x, const W& w);
    constexpr auto value() -> T;
};

template<class T, class W, std::weakly_increamentable O>
requires std::convertible_to<T, std::iter_value_t<O>>
class weighted_modes_of_sorted_accumulator
{
public:
    explicit weighted_modes_of_sorted_accumulator(size_t n, O it) noexcept;
    constexpr void operator() (const T& x, const W& w);
    constexpr auto value() -> O;
};

```

```

explicit mode_of_sorted_accumulator() noexcept;
explicit modes_of_sorted_accumulator(size_t n, O it) noexcept;
explicit weighted_mode_of_sorted_accumulator() noexcept;
explicit weighted_modes_of_sorted_accumulator(size_t n, O it) noexcept;

```

1. *Effects*: A (weighted) mode(s) accumulator object is constructed.
2. *Complexity*: Constant.

```
constexpr void operator() (const T& x);
```

1. *Effects*: The value of x is accumulated.
2. *Complexity*: Constant.

```
constexpr void operator() (const T& x, const W& w);
```

1. *Effects*: The value of x, weighted by w, is accumulated.
2. *Complexity*: Constant.

```
constexpr auto value() const -> T;
constexpr auto value() const -> O;
```

1. *Preconditions:* The (weighted) values of the associated range r (weighted by the corresponding values of the associated range w) have been accumulated, where r is a sorted range that has at least 1 value and the length of r is less than or equal to the length of w .
2. *Effects:* Any remaining computations relating to the (weighted) mode(s) are performed.
3. *Returns:* Any (weighted) mode of the values of the associated range r (weighted by the corresponding values of the associated range w) in the case of `mode_of_sorted_accumulator` and `weighted_mode_of_sorted_accumulator` and an output iterator past the last element copied to any n (weighted) modes in any order otherwise.
4. *Complexity:* Constant.

5.2.2 Covariance Accumulator Class Templates

```
template<class T1, class T2>
requires convertible_to<T1, T2> || convertible_to<T2, T1>
class covariance_accumulator
{
public:
    explicit constexpr covariance_accumulator(
        std::conditional<std::is_convertible_v<T1, T2>, T2, T1>::type ddof) noexcept;
    constexpr void operator()(const T1& x, const T2& y);
    constexpr auto value() -> std::conditional<std::is_convertible_v<T1, T2>, T2, T1>::type;
};

template<class T1, class T2, class W>
requires convertible_to<T1, T2> || convertible_to<T2, T1>
class weighted_population_covariance_accumulator
{
public:
    explicit constexpr weighted_population_covariance_accumulator() noexcept;
    constexpr void operator()(const T1& x, const T2& y, const W& w);
    constexpr auto value() -> std::conditional<std::is_convertible_v<T1, T2>, T2, T1>::type;
};

template<class T1, class T2, class W>
requires convertible_to<T1, T2> || convertible_to<T2, T1>
class weighted_sample_covariance_accumulator
{
public:
    explicit constexpr weighted_sample_covariance_accumulator() noexcept;
    constexpr void operator()(const T1& x, const T2& y, const W& w);
    constexpr auto value() -> std::conditional<std::is_convertible_v<T1, T2>, T2, T1>::type;
};

explicit constexpr covariance_accumulator(
    std::conditional<std::is_convertible_v<T1, T2>, T2, T1>::type ddof) noexcept;
explicit constexpr weighted_population_covariance_accumulator();
explicit constexpr weighted_sample_covariance_accumulator() noexcept;
```

1. *Effects:* A (weighted) covariance accumulator object is constructed.
2. *Complexity:* Constant.

```
constexpr void operator()(const T1& x, const T2& y);
```

1. *Effects:* The values of x and y are accumulated.
2. *Complexity:* Constant.

```
constexpr void operator() (const T1& x, const T2& y, const W& w);
```

1. *Effects*: The values of x and y , weighted by w , are accumulated.
2. *Complexity*: Constant.

```
std::conditional<std::is_convertible_v<T1, T2>, T2, T1>::type;
```

1. *Preconditions*: The (weighted) values of the associated range r (weighted by the corresponding values of the associated range w) have been accumulated, where r has at least 1 value and the length of r is less than or equal to the length of w .
2. *Effects*: Any remaining computations relating to the (weighted) covariance are performed.
3. *Returns*: The (weighted) covariance of the values of the associated range r (weighted by the corresponding values of the associated range w).
4. *Complexity*: Constant.

5.2.3 Correlation Accumulator Class Templates

```
template<class T1, class T2>
requires convertible_to<T1, T2> || convertible_to<T2, T1>
class correlation_accumulator
{
public:
    explicit constexpr correlation_accumulator(
        std::conditional<std::is_convertible_v<T1, T2>, T2, T1>::type ddof) noexcept;
    constexpr void operator() (const T1& x, const T2& y);
    constexpr auto value() -> std::conditional<std::is_convertible_v<T1, T2>, T2, T1>::type;
};
```

```
template<class T1, class T2, class W>
requires convertible_to<T1, T2> || convertible_to<T2, T1>
class weighted_population_correlation_accumulator
{
public:
    explicit constexpr weighted_population_correlation_accumulator() noexcept;
    constexpr void operator() (const T1& x, const T2& y, const W& w);
    constexpr auto value() -> std::conditional<std::is_convertible_v<T1, T2>, T2, T1>::type;
};
```

```
template<class T1, class T2, class W>
requires convertible_to<T1, T2> || convertible_to<T2, T1>
class weighted_sample_correlation_accumulator
{
public:
    explicit constexpr weighted_sample_correlation_accumulator() noexcept;
    constexpr void operator() (const T1& x, const T2& y, const W& w);
    constexpr auto value() -> std::conditional<std::is_convertible_v<T1, T2>, T2, T1>::type;
};
```

```
explicit correlation_accumulator(
    std::conditional<std::is_convertible_v<T1, T2>, T2, T1>::type ddof) noexcept;
explicit constexpr weighted_population_correlation_accumulator() noexcept;
explicit constexpr weighted_sample_correlation_accumulator() noexcept;
```

1. *Effects*: A (weighted) correlation accumulator object is constructed.

2. *Complexity*: Constant.

```
constexpr void operator() (const T& x, const T& y);
```

1. *Effects*: The values of x and y are accumulated.

2. *Complexity*: Constant.

```
constexpr void operator() (const T& x, const T& y, const W& w);
```

1. *Effects*: The values of x and y , weighted by w , are accumulated.

2. *Complexity*: Constant.

```
constexpr auto value() -> std::conditional<std::is_convertible_v<T1, T2>, T2, T1>::type;
```

1. *Preconditions*: The (weighted) values of the associated range r (weighted by the corresponding values of the associated range w) have been accumulated, where r has at least 1 value and the length of r is less than or equal to the length of w .

2. *Effects*: Any remaining computations relating to the (weighted) correlation are performed.

3. *Returns*: The (weighted) correlation of the values of the associated range r (weighted by the corresponding values of the associated range w).

4. *Complexity*: Constant.

5.2.4 Linear Regression Accumulator Class Templates

```
template<class T1, class T2>
requires convertible_to<T1, T2> || convertible_to<T2, T1>
class linear_regression_accumulator
{
public:
    explicit constexpr linear_regression_accumulator() noexcept;
    constexpr void operator() (const T1& x, const T2& y);
    constexpr auto value() -> linear_regression_result<
        typename std::conditional<std::is_convertible_v<T1, T2>, T2, T1>::type>;
};

template<class T1, class T2, class W>
requires convertible_to<T1, T2> || convertible_to<T2, T1>
class weighted_population_linear_regression_accumulator
{
public:
    explicit constexpr weighted_population_linear_regression_accumulator() noexcept;
    constexpr void operator() (const T1& x, const T2& y, const W& w);
    constexpr auto value() -> linear_regression_result<
        typename std::conditional<std::is_convertible_v<T1, T2>, T2, T1>::type>;
};

template<class T1, class T2, class W>
requires convertible_to<T1, T2> || convertible_to<T2, T1>
class weighted_sample_linear_regression_accumulator
{
public:
    explicit constexpr weighted_sample_linear_regression_accumulator() noexcept;
    constexpr void operator() (const T1& x, const T2& y, const W& w);
    constexpr auto value() -> linear_regression_result<
        typename std::conditional<std::is_convertible_v<T1, T2>, T2, T1>::type>;
};
```



```
explicit constexpr linear_regression_accumulator() noexcept;
explicit constexpr weighted_population_linear_regression_accumulator() noexcept;
explicit constexpr weighted_sample_linear_regression_accumulator() noexcept;
```

1. *Effects*: A (weighted) linear regression accumulator object is constructed.
2. *Complexity*: Constant.

```
constexpr void operator() (const T& x, const T& y);
```

1. *Effects*: The values of x and y are accumulated.
2. *Complexity*: Constant.

```
constexpr void operator() (const T& x, const T& y, const W& w);
```

1. *Effects*: The values of x and y , weighted by w , are accumulated.
2. *Complexity*: Constant.

```
constexpr auto value() -> linear_regression_result<
    typename std::conditional<std::is_convertible_v<T1, T2>, T2, T1>::type>;
```

1. *Preconditions*: The (weighted) values of the associated range r (weighted by the corresponding values of the associated range w) have been accumulated, where r has at least 1 value and the length of r is less than or equal to the length of w .
2. *Effects*: Any remaining computations relating to the (weighted) y -intercept and slope of the linear regression are performed.
3. *Returns*: `std::pair` with iterators to the (weighted) y -intercept and slope of the linear regression of the values of the associated range r (weighted by the corresponding values of the associated range w).
4. *Complexity*: Constant.

5.2.5 Accumulator Objects Accumulation Functions

```
template<ranges::input_range R1, ranges::input_range R2, class ...Accumulators>
constexpr void stats_accumulate(ranges::zip_view<R1, R2>&& r, Accumulators&& ... acc);
```

```
template<ranges::input_range R1,
    ranges::input_range R2,
    ranges::input_range W,
    class ...Accumulators>
constexpr void weighted_stats_accumulate(
    ranges::zip_view<R1, R2>&& r, W&& w, Accumulators&& ... acc);
```

```
template<class ExecutionPolicy,
    ranges::input_range R1,
    ranges::input_range R2,
    class ...Accumulators>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>>
void stats_accumulate(
    ExecutionPolicy&& policy, ranges::zip_view<R1, R2>&& r, Accumulators&& ... acc);
```

```
template<class ExecutionPolicy,
    ranges::input_range R1,
    ranges::input_range R2,
    ranges::input_range W,
```

```

class ...Accumulators>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>>
void weighted_stats_accumulate(
    ExecutionPolicy&& policy, ranges::zip_view<R1, R2>&& r, W&& w, Accumulators&& ... acc);

```

1. *Preconditions*: `r` and `w` are ranges of finite values, where `r` has at least 1 value and the length of `r` is less than or equal to the length of `w` and `acc` are valid accumulator objects.
2. *Effects*: The (weighted) statistics of the accumulator objects `acc` over the values of `r` (weighted by the corresponding values of the associated range `w`) are computed.
3. *Complexity*: Linear in `ranges::distance(r)`.

The preconditions of the **Accumulator Objects Accumulation Functions** section of P1708R6 [1] is to be updated to be

1. *Preconditions*: `r` and `w` are ranges of finite values, sorted if any of the accumulator objects of `acc` is a mode accumulator object, where the length of `r` is less than or equal to the length of `w`, `r` has at least 4 values if any of the accumulator objects of `acc` is a kurtosis accumulator object, `r` has at least 3 values if any of the accumulator objects of `acc` is a skewness accumulator object and `r` has at least 1 value otherwise, and `acc` are valid accumulator objects.

5.3 Freestanding Functions

If any of the values of the ranges `r` or `w` of the functions specified in this section is a NaN (Not a Number), ∞ or $-\infty$, or overflow or underflow occurs, which might even occur in the case of finite ranges of values, the function returns an unspecified value.

5.3.1 Freestanding Percentile Functions

```

template<ranges::sized_range R>
constexpr auto percentile_of_sorted(R&& r, double p) ->
    ranges::subrange<ranges::iterator_t<R>>;

template<ranges::sized_range R, ranges::input_range P, std::weakly_increamentable O>

/* ... requires ... */

constexpr auto percentiles_of_sorted(R&& r, P&& p, O it) -> O;

template<ranges::sized_range R>
constexpr auto median_of_sorted(R&& r) -> ranges::subrange<ranges::iterator_t<R>>;

template<ranges::sized_range R, ranges::input_range W>
constexpr auto weighted_median_of_sorted(R&& r, W&& w) ->
    ranges::subrange<ranges::iterator_t<R>>;

template<class ExecutionPolicy, ranges::sized_range R>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>>
constexpr auto percentile_of_sorted(ExecutionPolicy&& policy, R&& r, double p) ->
    ranges::subrange<ranges::iterator_t<R>>;

template<class ExecutionPolicy,
    ranges::sized_range R,
    ranges::input_range P,
    std::weakly_increamentable O>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>>

/* ... && ... */

constexpr auto percentiles_of_sorted(ExecutionPolicy&& policy, R&& r, P&& p, O it) -> O;

```

```

template<class ExecutionPolicy, ranges::sized_range R>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>>
constexpr auto median_of_sorted(ExecutionPolicy&& policy, R&& r) ->
    ranges::subrange<ranges::iterator_t<R>>;

template<class ExecutionPolicy, ranges::sized_range R, ranges::input_range W>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>>
constexpr auto weighted_median_of_sorted(ExecutionPolicy&& policy, R&& r, W&& w) ->
    ranges::subrange<ranges::iterator_t<R>>;

```

1. *Preconditions:* r and w are ranges of finite values, where r has at least 1 value and the length of r is less than or equal to the length of w and it is a valid iterator.
2. *Returns:* The (weighted) percentile(s) of the values of r (weighted by the corresponding values of the associated range w).
3. *Complexity:* Linear in n if r (and w) is or is derived from `random_access_range` and linear in `ranges::distance(r)` otherwise.

5.3.2 Freestanding Mode Functions

```

template<ranges::input_range R>
constexpr auto mode_of_sorted(R&& r) -> ranges::iterator_t<R>::value_type;

template<ranges::input_range R, ranges::input_range W>
constexpr auto weighted_mode_of_sorted(R&& r, W&& w) -> ranges::iterator_t<R>::value_type;

template<ranges::input_range R, std::weakly_increamentable O>
requires std::indirectly_copyable<ranges::iterator_t<R>, O>
constexpr auto modes_of_sorted(R&& r, size_t n, O it) -> O;

template<ranges::input_range R, ranges::input_range W, std::weakly_increamentable O>
requires std::indirectly_copyable<ranges::iterator_t<R>, O>
constexpr auto weighted_modes_of_sorted(R&& r, W&& w, size_t n, O it) -> O;

template<class ExecutionPolicy, ranges::input_range R>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>>
constexpr auto mode_of_sorted(ExecutionPolicy&& policy, R&& r) ->
    ranges::iterator_t<R>::value_type;

template<class ExecutionPolicy, ranges::input_range R, ranges::input_range W>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>>
constexpr auto weighted_mode_of_sorted(ExecutionPolicy&& policy, R&& r, W&& w) ->
    ranges::iterator_t<R>::value_type;

template<class ExecutionPolicy, ranges::input_range R, std::weakly_increamentable O>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>> &&
    std::indirectly_copyable<ranges::iterator_t<R>, O>
constexpr auto modes_of_sorted(ExecutionPolicy&& policy, R&& r, size_t n, O it) -> O;

template<class ExecutionPolicy,
    ranges::input_range R,
    ranges::input_range W,
    std::weakly_increamentable O>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>> &&
    std::indirectly_copyable<ranges::iterator_t<R>, O>
constexpr auto weighted_modes_of_sorted(
    ExecutionPolicy&& policy, R&& r, W&& w, size_t n, O it) -> O;

```

1. *Preconditions:* `r` and `w` are ranges of finite values, where `r` has at least 1 value and the length of `r` is less than or equal to the length of `w` and it is a valid iterator.
2. *Returns:* Any (weighted) mode of the values of the associated range `r` (weighted by the corresponding values of the associated range `w`) in the case of `mode_of_sorted_accumulator` and `weighted_mode_of_sorted_accumulator` and an output iterator past the last element copied to any `n` (weighted) modes in any order otherwise.
3. *Complexity:* Linear in `ranges::distance(r)`.

5.3.3 Freestanding Covariance Functions

```

template<ranges::input_range R1, ranges::input_range R2>
requires convertible_to<R1, R2> || convertible_to<R2, R1>
constexpr auto covariance(ranges::zip_view<R1, R2>&& r,
    typename std::conditional<
        std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
        std::iter_value_t<R1>,
        std::iter_value_t<R2>>::type ddof) ->
    std::conditional<
        std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
        std::iter_value_t<R1>,
        std::iter_value_t<R2>>::type;

template<ranges::input_range R1, ranges::input_range R2, ranges::input_range W>
requires convertible_to<R1, R2> || convertible_to<R2, R1>
constexpr auto weighted_population_covariance(ranges::zip_view<R1, R2>&& r, W&& w) ->
    std::conditional<std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
        std::iter_value_t<R1>,
        std::iter_value_t<R2>>::type;

template<ranges::input_range R1, ranges::input_range R2, ranges::input_range W>
requires convertible_to<R1, R2> || convertible_to<R2, R1>
constexpr auto weighted_sample_covariance(ranges::zip_view<R1, R2>&& r, W&& w) ->
    std::conditional<std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
        std::iter_value_t<R1>,
        std::iter_value_t<R2>>::type;

template<class ExecutionPolicy, ranges::input_range R1, ranges::input_range R2>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>> &&
    (convertible_to<R1, R2> || convertible_to<R2, R1>)
constexpr auto covariance(ExecutionPolicy&& policy,
    ranges::zip_view<R1, R2>&& r,
    typename std::conditional<std::is_convertible_v<
        std::iter_value_t<R1>, std::iter_value_t<R2>>,
        std::iter_value_t<R1>,
        std::iter_value_t<R2>>::type ddof) ->
    std::conditional<
        std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
        std::iter_value_t<R1>,
        std::iter_value_t<R2>>::type;

template<class ExecutionPolicy,
    ranges::input_range R1,
    ranges::input_range R2,
    ranges::input_range W>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>> &&
    (convertible_to<R1, R2> || convertible_to<R2, R1>)
constexpr auto weighted_population_covariance(
    ExecutionPolicy&& policy, ranges::zip_view<R1, R2>&& r, W&& w) ->

```

```

std::conditional<std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
std::iter_value_t<R1>,
std::iter_value_t<R2>>::type;

template<class ExecutionPolicy,
ranges::input_range R1,
ranges::input_range R2,
ranges::input_range W>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>> &&
(convertible_to<R1, R2> || convertible_to<R2, R1>)
constexpr auto weighted_sample_covariance(
ExecutionPolicy&& policy, ranges::zip_view<R1, R2>&& r, W&& w) ->
std::conditional<std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
std::iter_value_t<R1>,
std::iter_value_t<R2>>::type;

```

1. *Preconditions*: r and w are ranges of finite values, where r has at least 1 value and the length of r is less than or equal to the length of w .
2. *Returns*: The (weighted) covariance of the values of r (weighted by the corresponding values of the associated range w).
3. *Complexity*: Linear in `ranges::distance(r)`.

5.3.4 Freestanding Correlation Functions

```

template<ranges::input_range R1, ranges::input_range R2>
requires convertible_to<R1, R2> || convertible_to<R2, R1>
constexpr auto correlation(ranges::zip_view<R1, R2>&& r,
typename std::conditional<
std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
std::iter_value_t<R1>,
std::iter_value_t<R2>>::type ddof) ->
std::conditional<
std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
std::iter_value_t<R1>,
std::iter_value_t<R2>>::type;

template<ranges::input_range R1, ranges::input_range R2, ranges::input_range W>
requires convertible_to<R1, R2> || convertible_to<R2, R1>
constexpr auto weighted_population_correlation(ranges::zip_view<R1, R2>&& r, W&& w) ->
std::conditional<std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
std::iter_value_t<R1>,
std::iter_value_t<R2>>::type;

template<ranges::input_range R1, ranges::input_range R2, ranges::input_range W>
requires convertible_to<R1, R2> || convertible_to<R2, R1>
constexpr auto weighted_sample_correlation(ranges::zip_view<R1, R2>&& r, W&& w) ->
std::conditional<std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
std::iter_value_t<R1>,
std::iter_value_t<R2>>::type;

template<class ExecutionPolicy, ranges::input_range R1, ranges::input_range R2>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>> &&
(convertible_to<R1, R2> || convertible_to<R2, R1>)
constexpr auto correlation(ExecutionPolicy&& policy,
ranges::zip_view<R1, R2>&& r,
typename std::conditional<std::is_convertible_v<
std::iter_value_t<R1>, std::iter_value_t<R2>>,

```

```

    std::iter_value_t<R1>,
    std::iter_value_t<R2>>::type ddof) ->
    std::conditional<
        std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
        std::iter_value_t<R1>,
        std::iter_value_t<R2>>::type;

template<class ExecutionPolicy,
    ranges::input_range R1,
    ranges::input_range R2,
    ranges::input_range W>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>> &&
    (convertible_to<R1, R2> || convertible_to<R2, R1>)
constexpr auto weighted_population_correlation(
    ExecutionPolicy&& policy, ranges::zip_view<R1, R2>&& r, W&& w) ->
    std::conditional<std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
        std::iter_value_t<R1>,
        std::iter_value_t<R2>>::type;

template<class ExecutionPolicy,
    ranges::input_range R1,
    ranges::input_range R2,
    ranges::input_range W>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>> &&
    (convertible_to<R1, R2> || convertible_to<R2, R1>)
constexpr auto weighted_sample_correlation(
    ExecutionPolicy&& policy, ranges::zip_view<R1, R2>&& r, W&& w) ->
    std::conditional<std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
        std::iter_value_t<R1>,
        std::iter_value_t<R2>>::type;

```

1. *Preconditions*: `r` and `w` are ranges of finite values, where `r` has at least 1 value and the length of `r` is less than or equal to the length of `w`.
2. *Returns*: The (weighted) correlation of the values of `r` (weighted by the corresponding values of the associated range `w`).
3. *Complexity*: Linear in `ranges::distance(r)`.

5.3.5 Freestanding Linear Regression Functions

```

template<ranges::input_range R1, ranges::input_range R2>
requires convertible_to<R1, R2> || convertible_to<R2, R1>
constexpr auto linear_regression(ranges::zip_view<R1, R2>&& r) ->
    linear_regression_result<
        typename std::conditional<
            std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
            std::iter_value_t<R1>,
            std::iter_value_t<R2>>::type>;

template<ranges::input_range R1, ranges::input_range R2, ranges::input_range W>
requires convertible_to<R1, R2> || convertible_to<R2, R1>
constexpr auto weighted_population_linear_regression(
    ranges::zip_view<R1, R2>&& r, W&& w) ->
    linear_regression_result<
        typename std::conditional<
            std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
            std::iter_value_t<R1>,
            std::iter_value_t<R2>>::type>;

```

```

template<ranges::input_range R1, ranges::input_range R2, ranges::input_range W>
requires convertible_to<R1, R2> || convertible_to<R2, R1>
constexpr auto weighted_sample_linear_regression(
    ranges::zip_view<R1, R2>&& r, W&& w) ->
    linear_regression_result<
        typename std::conditional<
            std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
            std::iter_value_t<R1>,
            std::iter_value_t<R2>>::type>;

template<class ExecutionPolicy, ranges::input_range R1, ranges::input_range R2>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>> &&
    (convertible_to<R1, R2> || convertible_to<R2, R1>)
constexpr auto linear_regression(
    ExecutionPolicy&& policy, ranges::zip_view<R1, R2>&& r) ->
    linear_regression_result<
        typename std::conditional<
            std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
            std::iter_value_t<R1>,
            std::iter_value_t<R2>>::type>;

template<class ExecutionPolicy,
    ranges::input_range R1,
    ranges::input_range R2,
    ranges::input_range W>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>> &&
    (convertible_to<R1, R2> || convertible_to<R2, R1>)
auto weighted_population_linear_regression(
    ExecutionPolicy&& policy, ranges::zip_view<R1, R2>&& r, W&& w) ->
    linear_regression_result<
        typename std::conditional<
            std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
            std::iter_value_t<R1>,
            std::iter_value_t<R2>>::type>;

template<class ExecutionPolicy,
    ranges::input_range R1,
    ranges::input_range R2,
    ranges::input_range W>
requires std::is_execution_policy_v<std::remove_cvref_t<ExecutionPolicy>> &&
    (convertible_to<R1, R2> || convertible_to<R2, R1>)
auto weighted_sample_linear_regression(
    ExecutionPolicy&& policy, ranges::zip_view<R1, R2>&& r, W&& w) ->
    linear_regression_result<
        typename std::conditional<
            std::is_convertible_v<std::iter_value_t<R1>, std::iter_value_t<R2>>,
            std::iter_value_t<R1>,
            std::iter_value_t<R2>>::type>;

```

1. *Preconditions:* `r` and `w` are ranges of finite values, where `r` has at least 1 value and the length of `r` is less than or equal to the length of `w`.
2. *Returns:* `std::pair` with iterators to the (weighted) y -intercept and slope of the linear regression of the values of the associated range `r` (weighted by the corresponding values of the associated range `w`).
3. *Complexity:* Linear in `ranges::distance(r)`.

6 Acknowledgements

Michael Wong's work is made possible by Codeplay Software Ltd., ISO C++ Foundation, Khronos and the Standards Council of Canada. The authors of this proposal wish to further thank the members of SG19 for their contributions. Additional thanks are extended to Jolanta Opara, along with Axel Naumann of CERN.

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Appendix A Examples

The following example showcases the use of **percentile** functions.

```
std::vector<double> v = { 2.0, 3.0, 5.0, 7.0, 11.0, 13.0, 17.0, 19.0 };

std::vector<std::ranges::subrange<std::vector<double>::iterator>> percentiles_(3);
std::percentiles_of_sorted(v, std::vector<double>{0.25, 0.5, 0.75}, percentiles_.begin());
for (auto& p : percentiles_)
    std::cout << (*p.begin() + *std::prev(p.end())) /
        std::distance(p.begin(), p.end()) << " ";
```

The following example showcases the use of a **mode** function.

```
std::string text = "throughput";
std::ranges::sort(text);

std::vector<char> modes(text.size());
auto s_end = std::modes_of_sorted(text, modes.size(), modes.begin());

std::cout << "modes = ";
for (auto it = modes.cbegin(); it != s_end; ++it)
    std::cout << *it << " ";
```

The following example showcases the use of **modes** accumulator objects.

```
std::list<int> L = { 1, 2, 2, 2, 3, 3, 3 };

std::vector<int> modes(4);
std::mode_of_sorted_accumulator<int> m1;
std::modes_of_sorted_accumulator<int, std::vector<int>::iterator> m2(4, modes.begin());

std::stats_accumulate(L, m1, m2);

std::cout << "mode = " << m1.value();
std::cout << "\nmodes = ";
auto end = m2.value();
for (auto i = modes.cbegin(); i != end; i++)
    std::cout << *i << " ";
```

The following example showcases the use of **covariance**, **correlation** and **linear regression** functions.

```
std::vector<double> v1 = { 2.0, 3.0, 5.0, 7.0, 11.0, 13.0, 17.0, 19.0 };
std::vector<double> v2 = { -2.0, -3.0, -5.0, -7.0, -11.0, -13.0, -17.0, -19.0 };

std::cout << "covariance = " << std::covariance(std::views::zip(v1, v2), 0);

std::cout << "\ncorrelation = " << std::correlation(std::views::zip(v1, v2), 0);
```

```

auto [intercept, slope] = std::linear_regression(std::views::zip(v1, v2));
std::cout << "\nlinear regression = " << intercept << ", " << slope;

```

The following example showcases the use of **covariance**, **correlation** and **linear regression** accumulator objects.

```

/* custom accumulator */
class mean_squared_error_accumulator
{
public:
    constexpr mean_squared_error_accumulator() noexcept { MSE_ = 0; n_ = 0; }
    constexpr void operator()(double x, double y) { MSE_ += (x - y) * (x - y); n_++; }
    constexpr double value() const { return MSE_ / static_cast<double>(n_); }
private:
    double MSE_;
    size_t n_;
};

// ...

std::list<double> L1 = { 1, 2, 2, 3, 3 }, L2 = { 4, 7, 8, 9, 9 };

std::covariance_accumulator<double> covar(0);
std::correlation_accumulator<double> corr(0);
std::linear_regression_accumulator<double> reg;
mean_squared_error_accumulator mse;

std::stats_accumulate(std::views::zip(L1, L2), covar, corr, reg, mse);

std::cout << "covariance = " << covar.value();

std::cout << "\ncorrelation = " << corr.value();

auto [intercept, slope] = reg.value();
std::cout << "\nlinear regression = " << intercept << ", " << slope;

std::cout << "\nMSE = " << mse.value();

```